MIL-G-19769B(OS)
5 January 1976
SUPERSEDING
MIL-G-19769A(WP)
20 August 1957

MILITARY SPECIFICATION

GASKETS, SYNTHETIC RUBBER, OIL RESISTANT, SLIDE VALVE (FOR 21-INCH SUBMERGED TORPEDO TUBES)

This specification is approved for use by all departments and agencies of the Department of Defense.

1. SCOPE

*1.1 This specification covers submarine slide valve gaskets, using oil resistant synthetic rubber.

2. APPLICABLE DOCUMENTS

*2.1 The following documents of issues in effect on date of invitation for bids or request for proposals form a part of this specification to the extent specified herein. In the event of conflict between this specification and other documents referenced herein, requirements of this specification shall apply.

SPECIFICATION

Federal

*NN-P-530	Plywood, Flat Panel
PPP-B-591	Boxes, Fiberboard, Wood-Cleated
PPP-B-601	Boxes, Wood, Cleated Plywood
*PPP-B-621	Boxes, Wood, Nailed and Lock-Corner
*PPP-B-636	Boxes, Shipping, Fiberboard

FSC 9330

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Military

MIL-P-116	Preservation-Packaging, Methods of
*MIL-L-10547	Liners, Case, and Sheet, Overwrap, Water Vaporproof or Waterproof, Flexible

STANDARDS

Federal

FED-STD-601	Rubber: Sampling and Testing

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage

PUBLICATIONS

Naval Sea Systems Command

OP 1933	21-Inch Submerged Torpedo Tubes Mark 43 and 45 and Mods Description, Operation and Maintenance

(Copies of specifications, standards, drawings and publications requested by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Description. The gaskets to which this specification applies form the hydrostatic seal in the slide valve in hydraulic ejection submarine torpedo tubes. This valve is described in OP 1933 and controls the flow of sea water between the torpedo ejection pump and the individual torpedo tubes.

*3.2 General requirements.

3.2.1 Preproduction samples. Unless otherwise specified in the contract or order, preproduction samples of the gaskets shall be manufactured

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using the methods and procedures proposed for the production lot. These samples will be tested as specified herein and are for the purpose of determining that the production item meets the requirements of this specification and the design.

3.2.2 Material. The material shall be synthetic rubber oil-resistant compound of the butadiene-acrylonitril copolymer type equal to or better than that cited in 6.2 herein. It shall be oil and sea water resistant, durable, and shall not stick to metals nor cause corrosion or pitting of the metal parts with which it is assembled.

3.3 Form. The gaskets shall be furnished in the configuration and to the dimensions on the drawings specified in the contract or order.

3.4 Tolerances. Unless otherwise specified in the contract or order, tolerances shall be held within the limits specified on the drawings.

3.5 Physical requirements. The synthetic rubber material shall conform to the following physical requirements.

3.5.1 Before aging.

*3.5.1.1 Tensile strength shall be 2,500 pounds per square inch (psi), minimum.

*3.5.1.2 Ultimate elongation shall be 250 percent, minimum.

3.5.1.3 Hardness shall be as follows:

Plastometer, 0.125 inch		
Min.	Max.	Durometer
0.40	0.80	75 ± 5

3.5.2 After-oven aging. The synthetic rubber material, after having been subjected to the aging test specified in 4.3.4, shall meet the following requirements.

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3.5.2.1 Tensile strength. The tensile strength shall be not less than 80 percent of the tensile strength of unaged specimens.

3.5.2.2 Ultimate elongation. The ultimate elongation shall be not less than 60 percent of the ultimate elongation of unaged specimens.

3.5.2.3 Compression set. The compression set determined by the method specified in 4.3.3 shall be not more than 40 percent.

3.5.2.4 Hardness. The hardness determined by the method specified in 4.3.2 shall be not less than 70 durometer, nor more than 80.

3.5.3 Oil immersion. After being subjected to the conditions specified in 4.3.5, the synthetic rubber material shall meet the following requirements.

3.5.3.1 Swell. The change in volume of the specimen shall be not more than 3 percent when tested as specified in 4.3.6.

3.5.3.2 Tensile strength. The tensile strength shall be not less than 2,000 psi when tested as specified in 4.3.7.

*3.5.3.3 Ultimate elongation. The ultimate elongation shall be not less than 200 percent when tested as specified in 4.3.7.

3.6 Delamination. The synthetic rubber material shall show no evidence of separation into distinct layers or laminations when tested in accordance with 4.3.8.

3.7 Specify gravity. The specific gravity of the synthetic rubber material shall be not greater than 1.35 or less than 1.15 when determined as specified in 4.3.9.

3.8 Identification. A tag shall be placed on each individually wrapped gasket on which red symbols, such as diagonal stripes or rows of dots or their equivalent, shall be used to identify the gaskets. The date of marking shall be stamped thereon. The marking material shall be of a permanent type and shall not introduce any adverse affect causing changes to physcial characteristics.

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3.9 Workmanship. The gaskets shall be molded in a manner that will assure compliance with requirements of this specification. Attention shall be paid to neatness. The mold flash shall not exceed 0.003 inch in thickness at the point of contact with the gasket.

4. QUALITY ASSURANCE PROVISIONS

*4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements, as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

*4.1.1 Sampling. Unless otherwise specified, and when applicable, the sampling plans and procedures used in the determination of the acceptability of products submitted by a supplier for inspection shall be in accordance with the provisions set forth in MIL-STD-105.

*4.1.2 Preproduction sample. Unless otherwise specified for each contract or order, a representative sample of five gaskets shall be manufactured using the methods and procedures proposed for the production lot. These shall be submitted to the activity designated in the contract or purchase order for purposes of determining compliance with the requirements of the contract, specifications, and drawings. Further production of the item by the supplier prior to approval of the procuring activity or the completion of tests on preproduction sample shall be at the supplier's risk. Preproduction samples accepted will be supplied as part of the quantity specified by the contract or order.

4.1.3 Lot. For purposes of sampling, a "lot" shall consist of not more than 2,500 pounds of gaskets of the same design and dimensions produced in one plant under essentially the same conditions.

*4.1.4 Sampling procedure for lot acceptance inspection. For the inspection specified in 4.2.1, representative samples of each lot shall be selected in accordance with table I.

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Table I

SAMPLING PROCEDURE

Number of gaskets in lot	Number of gaskets to be selected
1 to 25	All
26 to 50	22
51 to 100	27
101 to 400	33
Over 400	36

*4.1.4.1 Lot size. As applied to inspection of units of product, the term "lot" shall mean "inspection lot," i.e., a collection of units of product submitted by a supplier for inspection. Unless otherwise specified, the number of units of product in inspection lots may differ from the quantity designated in the contract or order in a lot for production, shipment, or other purpose.

4.1.5 Sampling procedure for lot acceptance tests.

4.1.5.1 Sampling for tensile strength, elongation, and swell tests. The supplier shall furnish 3 sample pieces 6 inches long by 6 inches wide by 0.075 inch thick. These pieces shall be identical in composition, equivalent in cure, and prepared at the same time from material used in the lot offered for delivery.

4.1.5.2 Sampling for hardness and compression set tests. The supplier shall furnish 6 round sample pieces having a thickness of 0.500 ± 0.005 inch and cut with a die of 1.129 inches ± 0.10 inch in diameter. The sample pieces shall be identical in composition, equivalent in cure, and prepared at the same time from material used in the lot offered for delivery.

4.1.6 Sampling procedure for production tests. With the first lot of material offered for delivery under a contract or order and thenceforth, 4 sets of samples shall be furnished for every 10 lots which have passed inspection and test at the place of manufacture. Each set shall be prepared in accordance with 4.1.4 and furnished by the supplier. In no case shall a set of samples be taken from more than one lot. In addition, two 3-inch strips cut from two separate finished molded gaskets shall be forwarded to a designated Government laboratory to be checked for delamination.

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4.2 Inspection and tests.

4.2.1 Lot acceptance inspection (at place of manufacture). Each of the samples selected in accordance with 4.1.4 shall be subjected to surface inspection for workmanship, dimensions, and tolerances. Minor surface defects that do not affect the serviceability of the material shall not be cause for rejection. If any of the samples are found not to be in compliance with 3.3, 3.4, 3.8, and 3.9, the entire lot which it represents shall be rejected. A rejected lot may be resubmitted for inspection, but only after the manufacturer, subsequent to the first inspection, has inspected the lot 100 percent for the deficiency noted and has removed all nonconforming gaskets.

4.2.2 Lot acceptance tests.

*4.2.2.1 Place of tests. Lot acceptance tests shall be made at the place of manufacture if facilities are available. Otherwise, lot acceptance tests shall be made at a Government laboratory designated y the procuring activity, and samples therefor shall be forwarded to the designated activity.

*4.2.2.2 Test equipment. Test equipment or designs thereof shall be used in determining compliance with the acceptance crit eria, as specified herein.

4.2.2.3 Acceptance tests. The samples selected in accordance with 4.1.4.1 and 4.1.4.2 shall be subjected to the tests in 4.3.1 (initial tensile strength and ultimate elongation), 4.3.4 (tensile strength after aging), 4.3.3 (compression set after aging), 4.3.6 (swell), 4.3.2 (hardness before aging), and 4.3.4 (hardness after aging).

*4.2.2.4 Rejections. Failure of any one of the slide valve gasket samples representing a lot to meet any requirements of this specification, as determined by inspection and tests, shall be considered cause for rejection of the entire lot. At the option of the manufacturer, a rejected lot may be recured or otherwise reworked to remove the deficiencies which caused rejection and may then be resubmitted for Government inspection. In this case, the lot shall be resampled in accordance with 4.1.4. Additional samples shall be furnished by the supplier in accordance with 4.1.5, and both groups of samples shall be subjected to all inspections and tests specified in 4.2.1 and 4.2.2.3.

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4.2.4 Periodic production tests.

*4.2.4.1 Place of test. Periodic production tests shall be made at the place of manufacture if facilities are available. Otherwise, periodic production tests shall be made at a Government laboratory designated by the procuring activity and samples therefor shall be forwarded to the designated activity.

4.2.4.2 Tests. The samples selected in accordance with 4.1.6 shall be subjected individually to the tests in 4.3.7, 4.3.8, and 4.3.9.

4.2.4.3 Action in case of failure.

4.2.4.3.1 When tests are made at place of manufacture. If any one of the samples is found not to be in compliance with 3.5.3.1, 3.5.3.2, 3.5.3.3, 3.6, and 3.7, the lot which it represents shall be rejected. Additional samples shall be selected from each subsequent lot and shall be subjected to the test or tests wherein the failure occurred. A lot shall then be accepted only upon statisfactory completion of the test or tests by all the samples selected to represent the lot. This additional testing shall be discontinued when four successive lots have passed the test or tests.

*4.2.4.3.2 When tests are made at a Government laboratory. Acceptance of the first lot offered for delivery shall be withheld pending receipt of a satisfactory production check test report on the samples representing the lot. Except as hereinafter specified, acceptance or rejection of lots shall normally be on the basis of the sampling, inspection, and tests of 4.1.4, 4.2.1, and 4.2.2, and acceptance shall not be withheld pending receipt of test reports on production check test samples. However, upon receipt of an unsatisfactory test report on a production check test sample, additional samples shall be selected from every subsequent lot offered for delivery. The samples so selected shall be submitted to the tests wherein failure occurred. Lots shall then be accepted only upon receipt of a satisfactory test report on the samples so selected. This additional testing shall be discontinued and lot acceptance returned to the normal basis of 4.1.4, 4.2.1, and 4.2.2 when four successive lots have been accepted.

4.3 Test methods.

4.3.1 Tensile strength and elongation. Test specimens shall be in accordance with the requirements of methods 4111 and 4121 of FED-STD-601, using type III test specimens.

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4.3.2 Hardness. The hardness shall be determined by the test methods 3021 and 3031 of FED-STD-601. The specimens for both methods shall be 1/2 inch thick (±0.005 inch).

4.3.3 Aged compression set. The compression set test shall be in accordance with method 3311 of FED-STD-601, using a 25 percent deflection, with aging conditions of 4.3.4.

4.3.4 Aging. Specimens for tension test, hardness test, and for compression set test shall be given an accelerated aging test by subjecting them to dry circulating air at a temperature of 194° ± 2° F (90° ± 1.1° C) for 46 hours. Final determination of aged tension test specimens shall be made not less than 20 hours nor more than 48 hours after removal from the oven. Tension tests on unaged specimens shall be made immediately prior to, and on the same piece of apparatus, as the tension tests on the aged specimens.

4.3.5 Oil immersion. Specimens for swell test, tension test, and elongation test shall be immersed for 94 hours at 158° ± 3.6° F in a petroleum-base oil having the same viscosity, aniline point, and flash point as the No. 3 oil of method 6001 of FED-STD-601.

4.3.6 Swell.

4.3.6.1 Test specimens. The test specimen shall be 1 inch by 2 inches by 0.07 - 0.09 inch. Three specimens shall be used and the results averaged.

4.3.6.2 Procedure. The volume of each test specimen shall be measured in accordance with the procedure indicated in method 6231 of FED-STD-601 and 4.3.5.

4.3.7 Tension test after oil immersion. Three specimens shall be tested for tensile strength after being immersed in oil in accordance with 4.3.5 and method 6111 of FED-STD-601 except that the specimens shall be tested within 60 minutes from the time of removal from the immersion fluid. The tensile strength shall be determined on the basis of the swollen area as described in method 6111 of FED-STD-601.

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4.3.8 Delamination.

4.3.8.1 Method of test. A sample of strip rubber, 3 inches long and of the width and thickness furnished for delivery, shall be completely immersed in benzene, conforming to the characteristics specified in paragraph 4.3.8.2, for 24 hours. After removal from the benzene the strip shall be examined to determine evidence of separation into distinct layers or laminations by attempting to separate the laminations by hand.

4.3.8.2 Benzene. The benzene shall be a pure, water-white grade, free from suspended matter, conforming to the following characteristics:

Specific gravity at 60° F	0.875 to 0.886
Distillation limits:	
Initial boiling point	168° to 172° F
95 percent evaporated	174° F (max.)
Dry	176° F (max.)
Recovery	98 percent (min.)

*4.3.9 Specific gravity. The specific gravity shall be determined by using the hydrostatic weighing method 14011 of FED-STD-601, or other method satisfactory to the procuring activity.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.

5.1.1 Level A.

5.1.1.1 Cleaning. All gaskets shall be cleaned in accordance with method C1 of MIL-P-116.

5.1.1.2 Drying. After cleaning, all gaskets shall be dried in accordance with MIL-P-116, methods D1 or D4.

5.1.1.3 Unit packaging. Gaskets shall be packaged in accordance with method III of MIL-P-116. When practicable, each container shall contain an identical number of items and shall be of the same uniform size. In addition, all containers shall be new and shall be able to withstand storage, rehandling, and reshipment without the necessity of repacking.

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5.1.1.4 Intermediate packaging. Intermediate packaging shall be optional with the manufacturer.

5.2 Packing.

*5.2.1 Level A. Gaskets, packaged in accordance with 5.1.1.3 and 5.1.1.4 shall be packed in snug-fitting wood cleated fiberboard, cleated plywood, nailed wood, or exterior grade fiberboard boxes conforming to PPP-B-591, PPP-B-601 (overseas type), PPP-B-621, or PPP-B-636, respectively. Plywood shall conform to type 1 or 2, class 2 of NN-P-530. Containers shall be lined with a sealed waterproof case liner conforming to MIL-L-10547 and appendix thereto. Shipping containers shall be closed and strapped in accordance with the appendix of the applicable container specification. The gross weight of wood or wood-cleated boxes shall not exceed 70 pounds.

*5.3 Marking. In addition to the special markings required by the contract, order, unit packages, intermediate packages, and shipping containers shall be marked in accordance with the requirements of MIL-STD-129. The nomenclature shall be the exact nomenclature for the part or item that has been approved under the contract order.

6. NOTES

*6.1 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification

b. Form of material required (see 3.3)

c. Marking required (see 3.8)

d. Whether shipment is for immediate use, storage, or overseas shipment (see 5.1)

e. Special marking, if required (see 5.3).

*6.2 Requirements of this specification have been met by a compound of the following formula:

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Hycar 1042	100.00	Sulfur	1.50
Zinc oxide	5.00	Tricresyl phosphate	10.00
Spheron 9	80.67	Dibutyl phthalate	10.00
Cumar MH-2-1/2	6.66	Captax (MBT)	1.00
Agerite powder	1.00	Tetramethylthiuram disulfide	0.67
Steric acid	1.50		

*6.3 The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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OMB Approval
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DOCUMENT IDENTIFIER AND TITLE: MIL-G-19769B(OS), Gaskets, Synthetic Rubber, Oil Resistant, Slide Valve (For 21-Inch Submerged Torpedo Tubes)

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

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DATE

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